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Mechanism of CO₂ Enhanced Reverse Hydraulic Assisted Oil Displacement to Improve Recovery and Sequestration Ratio in Tight Reservoirs

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Abstract

Tight reservoirs are usually developed using horizontal well volume fracturing, but this method has problems such as rapid decline in production and low initial recovery. Therefore, there is an urgent need to explore new technologies to improve recovery. Hydraulic fracturing assisted oil displacement and CO₂ flooding have been proven to significantly improve the recovery of tight reservoirs, but these technologies are currently mostly applied independently, and no systematic integrated research has been conducted. The paper proposes for the first time the application of CO₂ enhanced reverse hydraulic fracturing assisted oil displacement to tight reservoirs. Through the use of magnetic resonance imaging equipment, CO₂ enhanced reverse hydraulic fracturing assisted oil displacements are conducted to quantify the contribution of recovery and CO₂ sequestration ratio at different stages, clarify the degree of mobilization of crude oil in microscopic pores, and reveal the evolution patterns of oil saturation and flow distance under different injection conditions. The results indicate that the oil recovery of CO₂ enhanced reverse hydraulic fracturing assisted oil displacement can reach 88.8%, with contributions of 12.3% from fracturing fluid imbibition, 73.2% from CO₂ flooding, and 3.3% from dissolved gas drive. The CO₂ sequestration ratio is 65.5%. Compared with conventional CO₂ flooding, under the same the injection volume, CO₂ enhanced reverse hydraulic fracturing assisted oil displacement has a slower forward propagation speed but a wider coverage range and higher CO₂ utilization efficiency. After CO₂ flooding, the depletion of driving energy and CO₂ breakthrough can easily cause residual oil enrichment at the production end. However, CO₂ enhanced reverse hydraulic fracturing assisted oil displacement improves the residual oil enrichment production end through synergistic mechanism such as fracturing energy replenishment, fracturing fluid imbibition, and high-pressure mixed-phase flooding. In addition, compared with water injection huff and puff, CO₂ injection huff and puff, and conventional CO₂ flooding, CO₂ enhanced reverse hydraulic fracturing assisted oil displacement increased the recovery by 16.7% to 75%. The results of this investigation provide a theoretical

basis for the field application of CO₂ enhanced reverse hydraulic fracturing assisted oil displacement in tight reservoirs.

Keywords: CO₂ enhanced reverse hydraulic fracturing assisted oil displacement, Recovery, Sequestration, Tight reservoirs

Introduction

With the continuous deepening of oil and gas exploration and development, the focus of China's oil and gas field development has shifted from traditional conventional oil and gas to unconventional sources such as low permeability, tight, and shale^[1]. These types of reservoirs are usually buried deeper, have greater heterogeneity, and face the challenge of poor development results in the early stages of development. Horizontal well volume fracturing technology has become a core method for the efficient development of tight and shale reservoirs^[2,3]. Although combining traditional water drive, chemical drive, and gas drive can improve recovery to a certain extent. There are still some technical issues that need to be addressed in the actual development process. Conventional chemical agents face the challenge of reduced concentration due to retention in pores, which weakens oil displacement efficiency. Conventional fracturing development is complex, costly, and has limited oil production effects^[4,5]. Therefore, a "pressure-injection-production" integrated enhanced oil recovery technology, also known as hydraulic fracturing assisted oil displacement, has been proposed. This is an unconventional oil and gas efficient development method that has been explored and refined in recent years based on actual development experience^[6,7]. Positive hydraulic fracturing oil displacement aims to solve the problems of high residual oil dispersion and ineffective water injection circulation, which is suitable for low-permeability reservoirs. Reverse hydraulic fracturing oil displacement is designed specifically for tight and shale reservoirs, aiming to solve the difficulties encountered in water injection development, such as high injection difficulty, high pressure gradient, and limited coverage^[8,9]. Compared with traditional fracturing oil recovery, hydraulic fracturing oil displacement has achieved significant oil recovery effect in oilfield applications. Positive hydraulic fracturing oil displacement involves creating fractures at the injection end at a pressure greater than the microfracture pressure of the reservoir. The fracturing fluid enters the formation pores along the fracture walls, collecting scattered residual oil at the production end^[10]. After the pressure drive injection is completed, the fractures close. At this point, due to the large amount of fracturing fluid/water entering the formation, while replenishing the formation energy, the wetting phase fluid imbibition replaces the non-wetting phase fluid^[11]. Subsequently, water injection development was performed in the injection well, promoting the further accumulation of fracturing fluid and crude oil toward the production well (Fig 1). The mechanism of positive hydraulic fracturing oil displacement to improve recovery includes (I) communicating microcracks to expand the range of propagation; (II) improving the pore structure; and (III) enhancing the oil recovery effect of fracturing through shut-in imbibition^[12]. Traditional hydraulic fracturing mainly promotes fracture extension to form single symmetrical long fractures. Water injection hydraulic fracturing oil displacement slows down fracture extension during fracturing, forming complex fractures that spread outward from the water injection well as the center. The characteristics of hydraulic fracturing oil displacement joint creation are short crack boundaries, small joint creation volume, which can avoid the chemical agent front breakthrough too early during the hydraulic fracturing oil displacement stage, thereby improving the propagation coefficient^[13,14]. Simultaneously, the hydraulic fracturing oil displacement water injection process produces a more balanced flow distribution, significantly increasing the wave propagation coefficient and constructing a high-pressure energy field across the entire reservoir. Compared with the conventional injection mode (26%), the wave propagation coefficient after pressure drive (77%) increased by approximately 51%^[15]. Pressure reduction and increased injection are effective ways to improve injection pressure and increase injection efficiency. By reducing injection pressure, energy consumption and operating costs are reduced, and the water absorption

capacity of the reservoir is improved by enhancing wettability^[16]. Continuous high-pressure injection causes fracturing of the formation, forming new fractures or expanding existing fractures, thereby improving the throat size and increasing the flow capacity. Imbibition has become an important method for improving recovery after fracturing tight reservoirs^[17]. During the shut-in imbibition process, pressure diffusion within the reservoir is promoted, improving the imbibition efficiency of the fracturing fluid. Reverse hydraulic fracturing oil displacement involves fracturing the formation with low-viscosity fracturing fluid from the wellhead (injection well closed, production well fractured), and the fracturing fluid enters the matrix along the fracture wall under displacement^[18]. After fracturing, sand slurry is injected and sand grains are added to support the fractures. After construction is completed, shut-in imbibition is performed to further enhance the interaction between the fracturing fluid and crude oil (both injection wells and production wells are shut down). After the shut-in period ends, water injection development resumes, and crude oil is collected at the production end through displacement (Fig 1, horizontal well at the production end). The mechanism of reverse pressure drive recovery includes (a) close cutting volume fracturing to increase controlled reserves; (b) large fluid injection to replenish formation energy; (c) surfactant-enhanced countercurrent imbibition. Close-spacing volume fracturing represents a significant breakthrough in the understanding and application of fracturing technology^[19]. Its core lies in optimizing fracture network structure and construction parameters, achieving efficient reservoir transformation. The main effects of close-spacing hydraulic fracturing are: disrupting reservoir structure, forming complex fracture networks to improve permeability; shortening the flow distance from the matrix to the fractures, reducing the flow displacement pressure difference; increasing single-well production and recovery, and expanding reserve utilization^[20]. Large-volume injection refers to the injection of a pre-treatment fluid containing surfactants prior to fracturing. By injecting a large volume of pre-treatment fluid, microfractures, induced fractures, and artificial fractures can be effectively connected, forming a complex pore-fracture network structure, which maximized the contact between fractures and the matrix. The mechanism of surfactant-enhanced countercurrent imbibition mainly includes: reducing the interfacial tension between oil and water, improving reservoir wettability, and promoting the formation of emulsified dispersion system for crude oil, thereby contributing to increased recovery in tight reservoirs^[21,22]. Cui et al. coupled crack propagation with a time-varying permeability model to study the dynamic characteristics of pressure drive production under different displacement^[23]. Huang et al. utilized physical simulation experiments to investigate the mechanism of hydraulic fracturing assisted oil displacement enhanced injection, clarifying the patterns of core fractur changes before and after hydraulic fracturing assisted oil displacement. Numerical simulation methods were utilized to characterize the dynamic diffusion mechanism at the leading edge of pressure drive, identifying the impact of injection-production well spacing, well network type, and water well injection volume on development effectiveness^[24]. Wang et al. analyzed the positive hydraulic fracturing assisted oil displacement mechanism through physical simulation experiments, clarifying the oil and water migration paths before and after pressure drive. The geological model was employed to analyze the types, rhythms, planar heterogeneity, and the impact of fractures on the hydraulic fracturing assisted oil displacement effect in low permeability reservoirs^[25]. Yu et al. established a coupled model of fracture extension seepage field and stress field based on the limited element discrete method, clarifying the mechanism and development effects of CO₂ pressure drive exploitation in low permeability heavy oil reservoirs^[26]. Chen et al. established constitutive equations between fluids and matrix based on Biot's line theory, characterized the dynamic expansion of pressure-driven injection water fractures, and revealed the mechanism of pressure-driven injection water to improve recovery^[27]. Li et al. conducted experimental investigations into the mechanism of flow, energy storage, oil washing, and return discharge of pressure driven agents, quantifying the contribution of recovery at different stages of the reverse hydraulic fracturing assisted oil displacement process^[28]. Wang et al. addressed the issue of limited historical data on reverse pressure drive by proposing an oil production enhancement prediction method based on AdaBoost-SVM. Which can effectively predict

the pressure drive effect of oil wells^[29]. Wang et al. conducted atmospheric pressure/high pressure dynamic adsorption experiments, analyzed the dynamic adsorption changes of pressure-driven agents on the core surface, and clarified the impact of high-pressure adsorption reduction on improving recovery^[30]. Qun et al. established a pressure-driven nonlinear seepage model that considers high-pressure drop adsorption, conducted an analysis of the effects of matrix permeability, fracture length, and initial pressure-driven agent concentration on the recharge effect^[31].

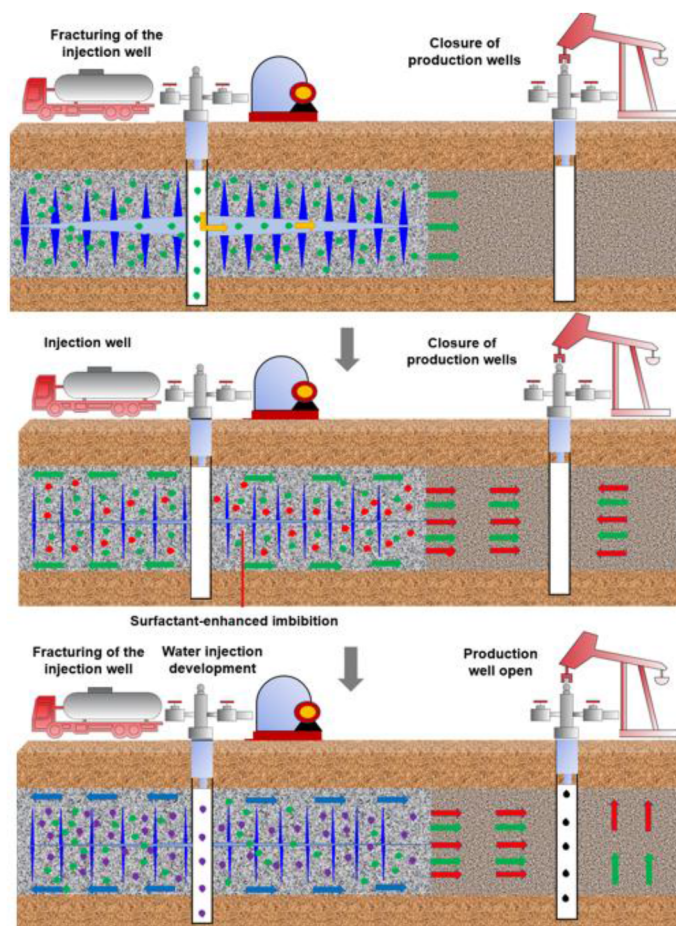


Figure 1—Schematic diagram of enhanced reverse hydraulic fracturing assisted oil displacement (production wells are horizontal well)

In summary, hydraulic fracturing assisted oil displacement has been proven to be an important means of improving reservoir recovery. However, due to limitations in reservoir characteristics and engineering implementation conditions, the application of reverse pressure drive in tight reservoirs is still not ideal, often resulting in problems such as insufficient displacement energy and low liquid production. Therefore, this paper proposes for the first time CO₂ enhanced reverse hydraulic assisted oil displacement technology (Fig 2). The contribution of recovery at different stages was quantified using nuclear resonance. Combining high-pressure mercury intrusion with nuclear magnetic resonance conversion, the degree of mobilization of crude oil in microporous space was quantitatively characterized. The saturation and distance evolution patterns during CO₂ enhanced reverse hydraulic assisted oil displacement are investigated based on GR-HSE. The distribution of residual oil in CO₂ flooding and CO₂ enhanced reverse hydraulic assisted oil displacement was determined using imaging technology. Finally, water injection huff and puff, CO₂ injection huff and puff development methods are compared, revealing the mechanism of CO₂ enhanced reverse hydraulic assisted

oil displacement to improve recovery. This research provides guidance for the field application of CO₂ enhanced reverse hydraulic assisted oil displacement in tight reservoirs.

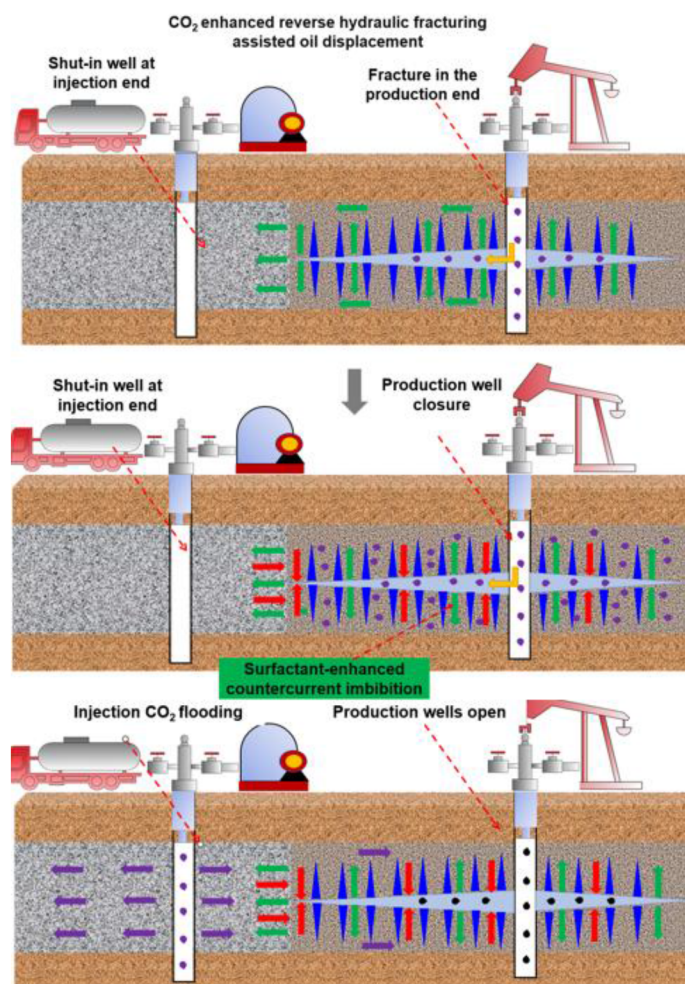


Figure 2—Schematic diagram of enhanced reverse hydraulic fracturing assisted oil displacement (production wells are horizontal well)

Materials and methods

The target block is the Chang 8 reservoir in Ordos Basin, with burial depth ranging from 1500 to 4000 m. The permeability ranges from $0.011 \times 10^{-3} \mu\text{m}^2$ to $0.8 \times 10^{-3} \mu\text{m}^2$, with an average value $0.25 \times 10^{-3} \mu\text{m}^2$, and the porosity distribution is 8.26%~11%, the mean value of the distribution of porosity was 9.857%. The porosity and permeability tester were employed to determine the parameters of the experimental cores, and the specific parameters are shown in Table 1.

Table 1—Core parameters

Number	Permeability/ $10^{-3} \mu\text{m}^2$	Porosity/%	Lengths/cm	Diameter/cm
A1	0.204	10.25	8.02	2.502
A2	0.213	9.97	8.06	2.492
A3	0.223	10.33	8.12	2.512
A4	0.209	10.06	7.96	2.517
A5	0.232	10.65	8.12	2.498

Fluids

Oil. The experimental crude oil was taken from the target block, the components of the crude oil were determined by gas chromatography, the results of which are shown in Fig 3. Under the condition of 25 °C and atmospheric pressure, the density of the crude oil was 0.823 g/cm³, and the viscosity was 8.23 mPa.s.

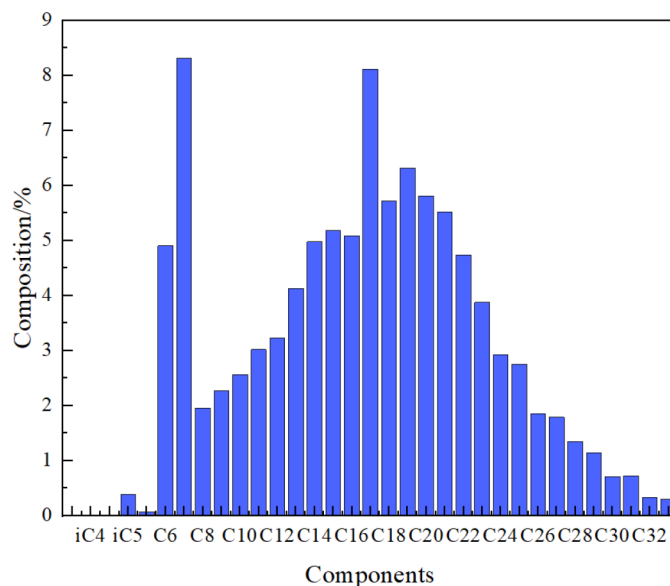


Figure 3—Component content of surface degassed crude oil

Formation water. The formation water type in the target block is CaCl₂ type with a PH of 6.1, the total mineralization is 35934 mg/L. The main ion types and components of formation water are illustrated in Table 2. The experimental water was prepared according to its composition and properties. In order to avoid the interference of hydrogen signal in formation water to the NMR test, manganese chloride was used to formulate the formation water in the experiment to ensure only crude oil signal during the experiment.

Table 2—Formation water properties

Fe ³⁺	Fe ²⁺	HCO ₃ ⁻	SO ₄ ²⁻	Cl ⁻	Mg ²⁺	Ca ²⁺	Na ⁺ , K ⁺	Total/(mg/L)
10	16	220	2578	19923	581	4397	8218	35934

Fracturing fluid. The type and concentration of the fracturing fluids utilized in the experiments were consistent with the actual applications in the field. The fluids involved during the experiments included CO₂, fracturing fluid and formation crude oil. In order to reduce the interference of the hydrogen signal in the fracturing fluid on the NMR test, manganese chloride (MnCl₂) was used to formulate the fracturing fluid in this experiment. After optimized treatment, neither the formation water nor fracturing fluid will interfere with the NMR test results.

Experimental design and scheme

Experimental setup. The CO₂ injection enhanced reverse hydraulic fracturing assisted oil displacement experimental setup consists of an NMR system and a core displacement system, which is illustrated in Fig 4. The online MRI is a MesoMR model, which consists of a computerized system, imaging, GR-HSE, and imaging reservoir with display system. The computer system is to process the core T₂, and GR-HSE data. The core replacement system includes injection pump, perimeter pressure pump, return pressure pump, and core gripper.

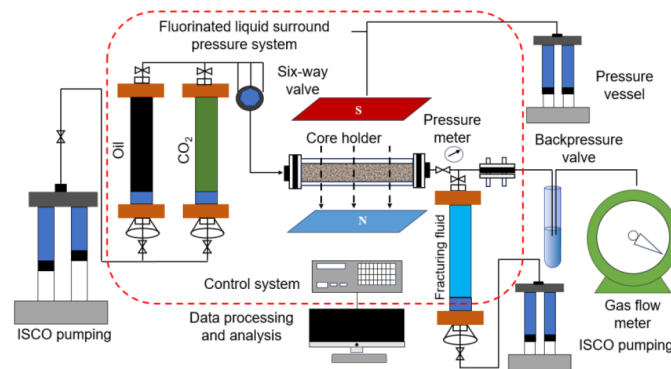


Figure 4—Experimental equipment

Experimental design and program. With the assistance of an on-line nuclear magnetic resonance analyzer, it was clarified that the fracturing fluid imbibition, CO₂ flooding and dissolved gas drive were contributing to the recovery. Combining pressurized mercury and NMR pore structure conversion, quantifying the mobilization pattern of crude oil in the microscopic pore space at each stage. The signal intensities of different interfaces in the core are measured by nuclear magnetic resonance pulse sequence (GR-HSE), which reveals the front migration characteristics of the fracturing fluid countercurrent imbibition and CO₂ oil flooding stage, and clarifies the evolution trend of injection saturation and distance. At the same time, spin echo imaging was used to analyze the residual oil characteristics inside the core. In order to accurately simulate the development process of hydraulic fracturing assisted oil displacement in the field, this investigation combines the parameters related to hydraulic fracturing assisted oil displacement in the field, according to the similarity principle to conduct the experimental design. The design theory of the similarity principle lies in the fact that the gradient of replacement pressure inside the core is equal to the target layer in the field. The specific equivalent method includes: (1) the pressure difference between the fracturing pressure applied to the formation and the target reservoir pressure, which is defined as the actual replacement pressure difference. (2) the ratio of the experimental core length of the actual seepage distance of the fracturing fluid in the reservoir is adopted as the similarity ratio. (3) the actual replacement pressure differential is converted from the similarity ratio, which determiner the required establishment of replacement pressure differential at both ends of the core. In the experiment, for reducing the error, four cores with similar permeability and pore structure were selected, and the pore distribution of the experimental core is shown in Fig 5. The specific experimental steps are as follows.

1. According to petroleum industry standards, the core samples are pretreated, then placed in an oven for drying. The dried cores are subsequently stored in a desiccator.
2. The dried core samples were placed in a vacuum chamber and vacuumed for seven days. After vacuuming, the experimental cores were pressurized and saturated with crude oil using a displacement pump. Upon completion of saturation, the cores were subjected to an aging process.
3. The replacement system was turned on raise the experimental temperature to predetermined conditions, while the CO₂ containing the intermediate vessel was preheated; the aging-treated experimental cores were placed in the NMR core holder.
4. After the temperature rise is completed, the experimental core is injected with oil and pressurized, increasing the pressure of the replacement system to a predetermined range. During the pressurization process, the fluoride liquid containment system was adjusted to ensure that the containment pressure was higher than the replacement pressure by 5 MPa. After stabilizing the liquid out of the production, oil injection was stopped, while the experimental core was processed with GR-HSE, NMR, and imaging scans, identifying different pore saturation and different cross-sectional hydrogen signal values for the initial state of the core.

5. Close the back pressure, ensuring that the pressure of the back pressure value will not decrease during the injection of fracturing fluid at the production end. Open the production end injection valve to inject 0.1 PV of fracturing fluid into the experimental core, and when the predetermined pressure is reached, simulated the stage of imbibition in the shut-in after hydraulic fracturing assisted oil displacement.
6. The experimental cores were scanned by NMR at the end of the shut-in to obtain T_2 curves under the shut-in imbibition conditions.
7. After the shut-in imbibition phase, the injection end was turned on for CO_2 flooding. During the displacement process, NMR scans were performed under different injection volume conditions to obtain T_2 spectra, GR-HSE data, and core imaging results.
8. When the pressure is reduced to the bubble point pressure and the NMR curve does not change, adjust the backpressure valve to gradually reduce to atmospheric pressure and, when the extracted end is no longer out of oil to stop the experiment.
9. At the end of the experiment the recovery was determined from the variation of the NMR curve, and the front migration pattern was determined from the variation of the saturation of the core section.
10. The CO_2 injection enhanced reverse hydraulic fracturing assisted oil displacement sequestration ratio was calculated, according to the change of CO_2 injection volume before and after the experiment.

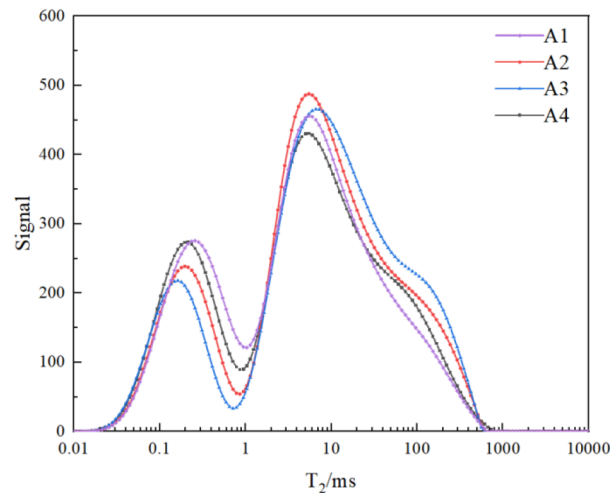


Figure 5—Core saturated oil NMR curves

Table 3—Experimental program

Number	Permeability/ $10^{-3}\mu\text{m}^2$	Porosity/%	Experimental content
A1	0.204	10.25	CO_2 enhanced reverse hydraulic fracturing assisted oil displacement
A2	0.213	9.97	Water injection huff and puff
A3	0.223	10.33	CO_2 puff and huff
A4	0.232	10.65	CO_2 flooding

Results and discussion

Enhanced recovery mechanisms

Recovery and microporous crude oil mobilization. The evolution curves of NMR signals during CO_2 enhanced reverse hydraulic assisted oil displacement in tight cores are shown in Fig 6, including the initial state, the imbibition stage of the fracturing fluid, the CO_2 flooding stage and the dissolved gas drive stage.

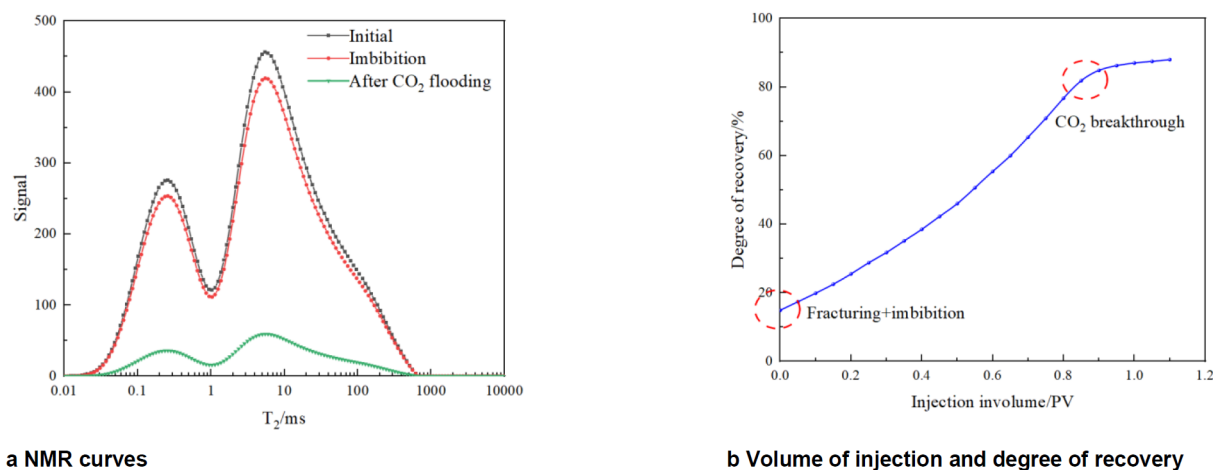


Figure 6—CO₂ injection enhanced reverse hydraulic fracturing assisted oil displacement NMR and recovery

Fracturing fluid imbibition contribution to recovery was determined by initial state versus NMR area variation after 12 h of shut-in imbibition. The overall recovery was determined by the area difference between the initial state and after CO₂ flooding. The ratio of the difference between the area of imbibition and CO₂ flooding to the initial area is the recovery contribution of the CO₂ flooding stage. At the end of the CO₂ flooding experiment, the system pressure is lowered to below the bubble point pressure, at which time the dissolved CO₂ in the crude resolves and secondary flooding occurs. The ratio of the area difference between the bubble point pressure and the end of CO₂ flooding to the initial area is the dissolved gas drive recovery. The findings indicated that the recovery of CO₂ enhanced reverse hydraulic assisted oil displacement from tight core injection was 88.7%. The fracturing fluid shut-in imbibition contribution was 12.2%, the CO₂ flooding stage recovery contribution was 73.2%. The contribution of recovery from dissolved gas drive was 3.3%. CO₂ enhanced reverse hydraulic assisted oil displacement achieves efficient mobilization of tight reservoirs by constructing a synergistic three-stage process: fracturing at the production end, fracturing fluid shut-in imbibition, and high-pressure CO₂ mixed-phase flooding.

The mechanism of CO₂ enhanced reverse hydraulic assisted oil displacement technology to improve recovery includes: (1) In the reverse hydraulic assisted oil displacement stage, fracturing is simulated by injecting fracturing fluid into the production end, which pushes the fracturing fluid into the matrix under the effect of differential pressure, prompting the capillary force to be converted from a resistance to a driving force, which enhances the fluid interaction between the matrix and the fracture. After the shut-in is finished, the fracturing fluid is not returned and the high-pressure environment of the core is maintained, which is conducive to the subsequent CO₂ mixed-phase flooding, preventing the early breakthrough of CO₂ and securing the stability of the drive process. (2) In the stage of shut-in imbibition, the fracturing fluid slowly enters the matrix under the effect of capillary force and pore pressure difference, improving the pore wettability, reducing the crude oil flow resistance, and providing an effective contact area for subsequent CO₂ flooding. (3) In the CO₂ flooding stage, CO₂ forms mixed-phase drive with crude oil under high-pressure conditions, which significantly reduces interfacial tension and capillary confinement, prompting CO₂ penetrate deep into the nan-micropores and effectively drive off the residual oil. By stabilizing the mixed-phase region, the timing of CO₂ breakthrough is delayed, thus improving the wave efficiency. Meanwhile, under high-pressure mixed-phase conditions, CO₂ has excellent solubility, which can reduce viscosity and enhance crude oil mobility, further enhancing the effect of displacement. (4) In the dissolved gas drive stage, at the end of CO₂ flooding, the system pressure is reduced to below the bubble point pressure, and the dissolved CO₂ in the crude oil precipitates out, providing gas expansion power to mobilize some of the residual oil trapped in the deeper part of the matrix, further enhancing the recovery.

The different pore space crude oil utilization is illustrated in Fig 7. The recovery degree of micropores (0.01~1 μm) is 12.2%, the recovery of mesopores 32.5% (1~10 μm), and the recovery degree of macropores (more than 10 μm) is 44%. The degree of recovery increased significantly with the larger pore size. CO₂ enhanced reverse hydraulic assisted oil displacement improved crude oil recovery at different pore scales, compared with conventional development methods. The largest increase in the degree of microporous mobilization, while the smallest increase in the microporous. During the initial stage of CO₂ enhanced reverse hydraulic fracturing oil displacement, the fracturing fluid slowly penetrates into micropores under the combined action of capillary force and pressure difference, improving the wettability of the pores and promoting the mobilization of crude oil in the microporous. In the later CO₂ flooding stage, since the resistance of micropores is relatively strong, CO₂ diffuses into the deep parts of the pores at a relatively slow rate, reducing the viscosity of crude oil and interfacial tension, thereby promoting low but effective utilization of microporous spaces. For mesopores, the shut-in stage at the beginning of reverse hydraulic fracturing assisted oil displacement promotes sufficient contact between the fracturing fluid and pores, improves wettability, and significantly mobilizes the crude oil in the mesopores. Upon entering the CO₂ flooding stage, CO₂ can rapidly diffuse to the pore scale under the influence of pore structure and flow resistance, forming a near-miscible or miscible state, reducing the viscosity of crude oil, and thereby improving the recovery efficiency of mesopores. At both the injection and production ends, CO₂ promotes oil and gas flow by improving wettability and reducing flow resistance. At the production end in particular, CO₂ further drives residual crude oil toward the production end after imbibition by reducing capillary confinement, thus effectively improving recovery. During the initial stage of reverse hydraulic fracturing assisted oil displacement, the fracturing fluid enters the macropores at a relatively fast rate, increasing the utilization of crude oil in the macropores and maintaining a high pore pressure. In the CO₂ injection stage, CO₂ preferentially enters the macropores, performing deep displacement and sweeping, thereby maximizing the recovery of the macropores.

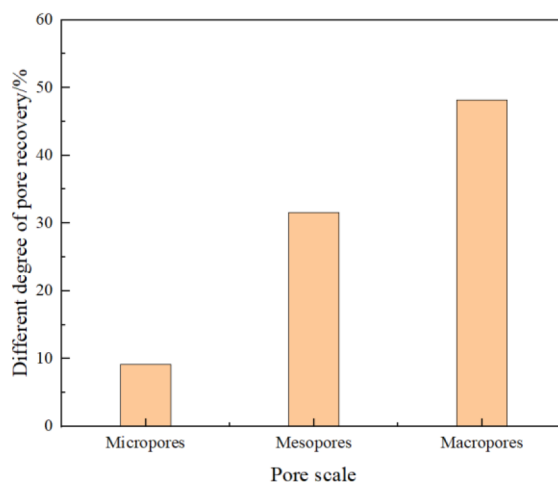


Figure 7—Degree of microscopic crude oil mobilization in different pore sizes

Saturation and distance. Figure 8 illustrates the evolution curve of oil saturation with distance during the CO₂ enhanced reverse hydraulic fracturing oil displacement process. From the injection end (0 mm) to the production end (80 mm), the residual oil saturation of the core increases gradually. As the injection volume increases, the CO₂ front gradually advances from the injection end to the production end. Compared with conventional CO₂ flooding, the CO₂ enhanced reverse hydraulic assisted oil displacement CO₂ front advances more slowly. The front part of the core (0~40mm) has a significant decrease in crude oil saturation and sufficient CO₂ flooding; the middle part (40~65 mm) shows a slow decrease in saturation, indicating

a displacement transition zone. There is a notice depression in the latter part (60~80 mm), where local enhanced mobilization occurs due to the combined effects of shut-in imbibition and CO₂ diffusion flooding, manifesting as a "low saturation plateau". During the shut-in phase, fracturing fluid imbibition into the matrix in the reverse direction, changing the wettability. In the later stages, CO₂ diffuses into the deeper pores and forms a miscible phase with the crude oil. Combined with dissolution, viscosity reduction, and expansion, this increases the mobilization of crude oil at the production end, forming a characteristic "distal mobilization zone." The imaging results before and after the experiment demonstrate the changes in oil saturation within the core (Fig 9, red indicates oil). Under the initial oil saturation conditions of the core, the oil phase was evenly distributed. However, after fracturing shut-in and CO₂ flooding, area of decreased saturation appeared in the front, middle, and the end of the core. Compared with conventional CO₂ flooding, the CO₂ enhanced reverse hydraulic assisted oil displacement method significantly reduced the saturation at the production end. This method effectively solved the problem of residual oil enrichment at the production end through the synergistic effect of fracturing fluid imbibition and CO₂ flooding.

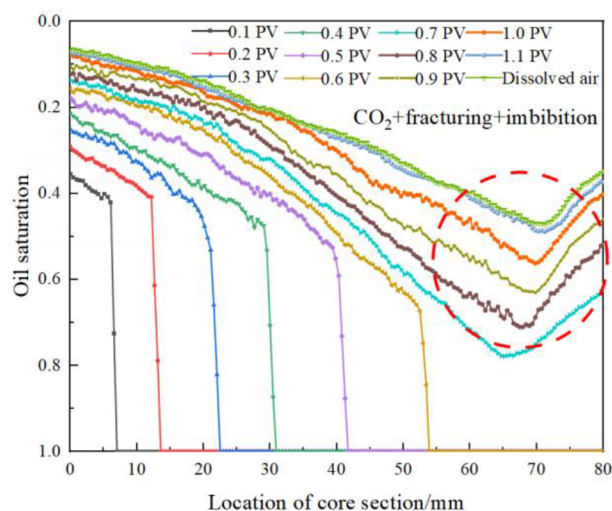


Figure 8—Variation of saturation and distance with different CO₂ injection volumes

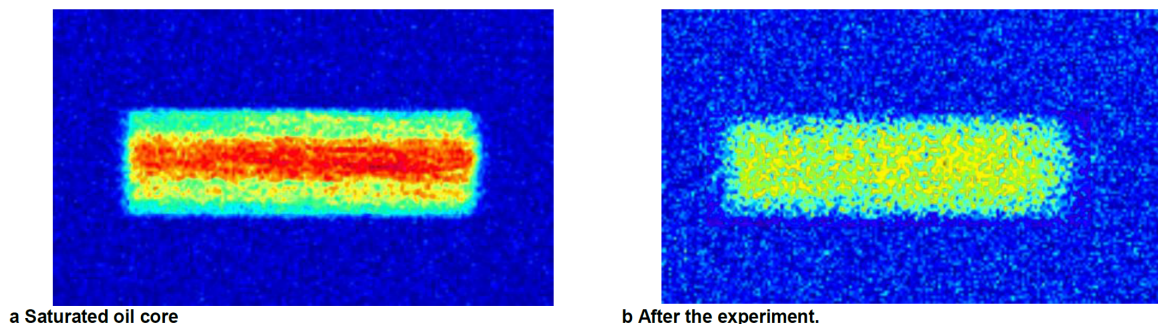


Figure 9—CO₂ injection enhanced reverse hydraulic assisted oil displacement magnetic resonance imaging

CO₂ sequestration. Figure 10 Comparison of the variation pattern of CO₂ sequestration ratio between conventional CO₂ flooding and CO₂ enhanced reverse hydraulic fracturing assisted oil displacement under different injection volume conditions. Experimental results indicate that, under the same injected volume fraction, CO₂ enhanced reverse hydraulic assisted oil displacement achieves a consistently higher sequestration effect. For both approaches, the sequestration ratio can be maintained close to 100% at the initial stage (0~0.7PV), indicating that the breakthrough has not yet occurred at the stage. After the breakthrough, the conventional CO₂ flooding sequestration ration decreased rapidly, while the enhanced

reverse hydraulic fracturing assisted oil displacement decreased more slowly, significantly delaying the CO₂ breakthrough time, and thus the method performed better in term of sequestration stability. The mechanism of CO₂ enhanced reverse hydraulic fracturing assisted oil displacement to improve the sequestration ratio includes: (1) The high-pressure shut-in stage to enhance the ability of mixed-phase drive, with no return drainage after hydraulic fracturing assisted oil displacement. Under high-pressure conditions, the CO₂ is retained in the core and core oil in a dissolved or mixed-phase state, which improves the stability of storage. (2) Imbibition releases microporous space, the phenomenon of imbibition occurring between the fracturing fluid and the crude oil during the shui-in contributes to the release of pore space, which can be utilized by sequestration, and the sequestration volume provided by the subsequent CO₂ injection. (3) Reverse hydraulic fracturing assisted oil displacement to promote CO₂ deep into the difficult to utilize area. In the later stage, CO₂ flooding is implemented from the injection end, and on the basis of the fracture formed at the production end with pressure conduction channels, the reverse pressure difference promotes CO₂ penetration into the deeper part of the matrix and low permeability area, effectively expanding the scope of drive and sequestration. CO₂ enhanced reverse hydraulic fracturing assisted oil displacement has delayed CO₂ breakthrough, and enhanced its retention capacity in microporosity, through a synergistic process of hydraulic fracturing assisted oil displacement at the production end, shui-in imbibition, and CO₂ drive at the injection end. This demonstrates higher sequestration efficiency than conventional CO₂ flooding, which is suitable for the development of CO₂ sequestration in tight reservoirs.

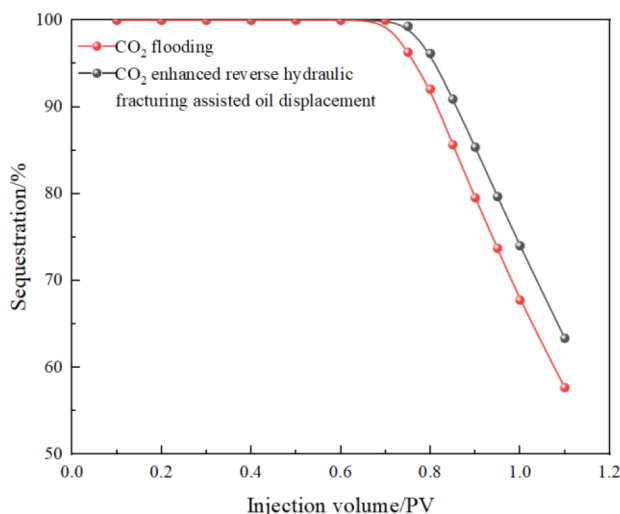


Figure 10—CO₂ sequestration ratio under different injection volumes

Comparison development methods

Water injection huff and puff. Figure 11a shows the NMR curves before and after water injection huff and puff. Figure 11b illustrates the difference in the degree of crude oil mobilization between water injection and CO₂ enhanced reverse hydraulic assisted oil displacement. The research results indicate that the water injection huff and puff is 14.6%. The recovery degree in micropores is 1.7%, in mesopores is 4.2%, and in macropores is 8.2%. Compared to the recovery of water injection huff and puff recovery and microporous crude oil utilization, CO₂ enhanced reverse hydraulic assisted oil displacement improves the recovery by 74.2%, with increases of 10.5% in micropores, 28.3% in mesopores, and 35.5% in macropores. Water injection huff and puff relies on water displacement to drive crude oil, but due to the low wave efficiency and large capillary confinement of water, it has poor mobilization effects in micropores and mesopores, and the recovery gradually decreases. In contrast, CO₂ enhanced reverse hydraulic assisted oil displacement

significantly improves the mobilization of crude oil through multiple mechanisms such as CO₂ miscible flooding, dissolution and viscosity reduction, improved wettability and fracturing fluid imbibition.

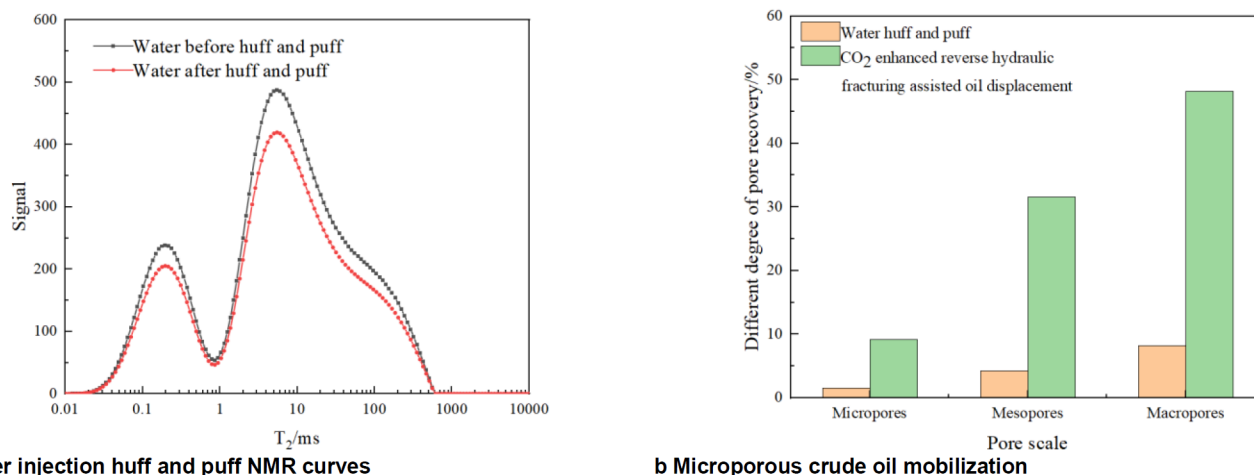


Figure 11—Water injection huff and puff NMR curves and microporous crude oil mobilization

CO₂ huff and puff. Figure 12a shows the NMR curves before and after CO₂ huff and puff injection tight cores, while figure 12b shows the degree of micropores crude oil mobilization under CO₂ huff and puff, water injection huff and puff, and CO₂ enhanced reverse hydraulic assisted oil displacement. The results indicate that the recovery degree of CO₂ huff and puff is 18.8%, with the recovery of micropores at 3.2%, mesopores at 5.9%, and macropores at 10.2%. Compared to CO₂ huff and puff, the recovery degree of CO₂ enhanced reverse hydraulic assisted oil displacement increases by 60%. The increases in micropores, mesopores, and macropores are 9%, 26.6%, and 33.8%. Compared to water injection huff and puff, the recovery degree of CO₂ huff and puff increases by 4.2%, with increases of 1.5%, 1.7%, and 2.0% in micropores, mesopores, and macropores, with the largest increase observed in macropores. CO₂ huffing and puffing mainly relies on the action of CO₂ dissolving crude oil. By injecting CO₂ into contact with crude oil for a certain period of time, the viscosity of crude oil is reduced, thereby promoting mobilization. However, due to its low propagation efficiency and difficulty in penetrating deep pores, its mobilization effect in complex reservoirs is limited.

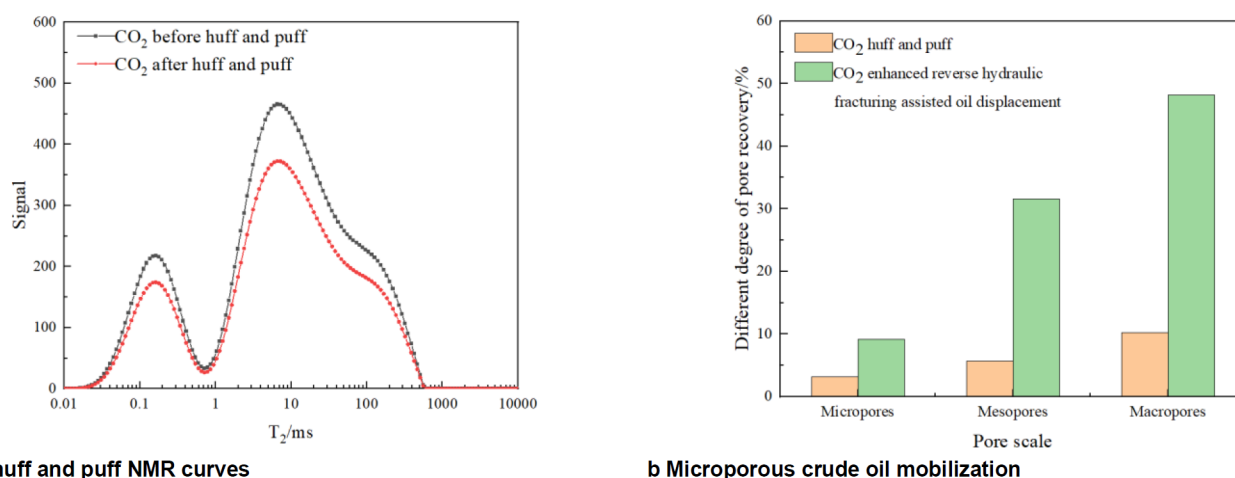


Figure 12—CO₂ huff and puff NMR curves and microporous crude oil mobilization

CO₂ flooding. To compare the effects of CO₂ flooding and CO₂ enhanced reverse hydraulic assisted oil displacement, experiments were conducted on CO₂ flooding and CO₂ enhanced reverse hydraulic assisted oil displacement. Figure 13a shows the nuclear magnetic resonance curves of tight rock cores before and after the CO₂ flooding experiment, 13b shows the curve of CO₂ flooding injection volume and recovery rate, and 13c shows the degree of crude oil mobilization in micro-pores during CO₂ flooding and CO₂ enhanced reverse hydraulic assisted oil displacement. With increasing injection volume, the degree of core recovery increases. Compared with CO₂ enhanced reverse hydraulic assisted oil displacement, when the injection volume is 0 PV, the CO₂ flooding recovery is 0%, while the recovery of CO₂ enhanced reverse hydraulic assisted oil displacement is 12.3% under the effect of fracturing shut-in imbibition. Under the same injection volume conditions, the CO₂ flooding oil recovery is 72.1%, with the recovery of micropores at 8.5%, mesopores at 26.9%, and macropores at 36.2%. The recovery increases CO₂ for enhanced reverse hydraulic assisted oil displacement in different pore types are 3.7%, 5.6%, and 7.3%, with the largest increase for macropores. Compared to water injection huff and puff, CO₂ injection increases the recovery by 61%, with recovery increases of 6.8% (micropores), 22.7% (mesopores), and 28.5% (macropores). Compared to CO₂ huff and puff, CO₂ flooding increases the recovery by 53.3%, with recovery increases of 5.3%, 21%, and 26.5% in different pore types. CO₂ enhanced reverse hydraulic assisted oil displacement significantly improves the degree of crude oil recovery through the synergistic action of multiple mechanisms, especially in reservoirs with complex pore structures. Although CO₂ displacement can reduce crude oil viscosity and improve fluidity, its wave effect is low and the wave range is limited.

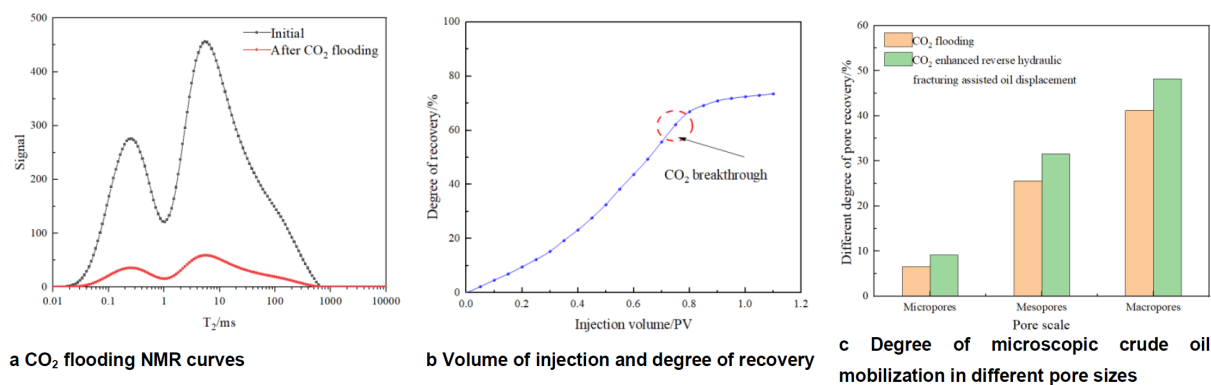


Figure 13—CO₂ flooding NMR and recovery

The GR-HSE magnetic resonance results demonstrate that as the CO₂ injection volume increases (Fig 14), the displacement front gradually advances toward the production end. In the initial stage, the front advances slowly, and the decrease in oil saturation within the core is small. In the middle stage of displacement, the front advances more rapidly, and oil saturation decreases significantly. In the late stage of displacement (after CO₂ breakthrough), both the front advancement speed and the rate of decrease in saturation tend to slow down. From the injection end (0 mm) to the production end (80 mm), the oil saturation decreases overall. The most pronounced decrease in saturation occurs in the 0–30 mm section, indicating the best CO₂ flooding effect; the flooding effect weakens in the 30–60 mm section, with a higher residual oil saturation; and in the 60–80 mm section, the CO₂ flooding effect further weakens, forming a distinct residual oil enrichment zone at the production end. The reason for residual oil accumulation is that during the later stages of CO₂ flooding, the pressure gradient significantly weakens, leading to insufficient driving force. Meanwhile, after CO₂ breakthrough, it advances along preferred pathways, bypassing areas outside the main flowline or low-permeability zones, resulting in some reservoirs not being effectively mobilized. Additionally, due to factors such as capillary forces and interfacial tension, crude oil in microporous spaces is difficult to be carried out by CO₂, further exacerbating residual oil retention in this area. This phenomenon is consistent with nuclear

magnetic resonance imaging results (Fig 15), both of which indicate the presence of distinct residual oil enrichment zones at the production end following CO₂ flooding.

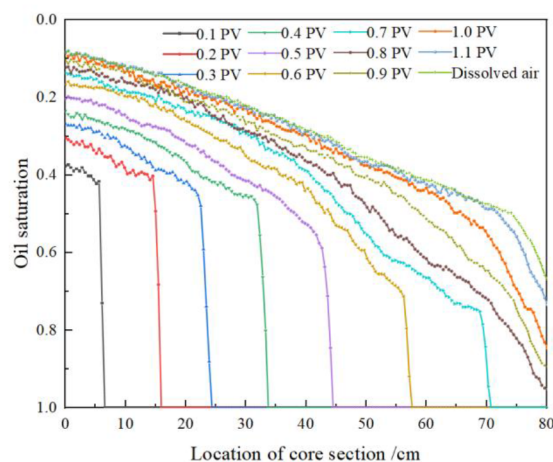


Figure 14—Variation of saturation and distance with different CO₂ injection volumes

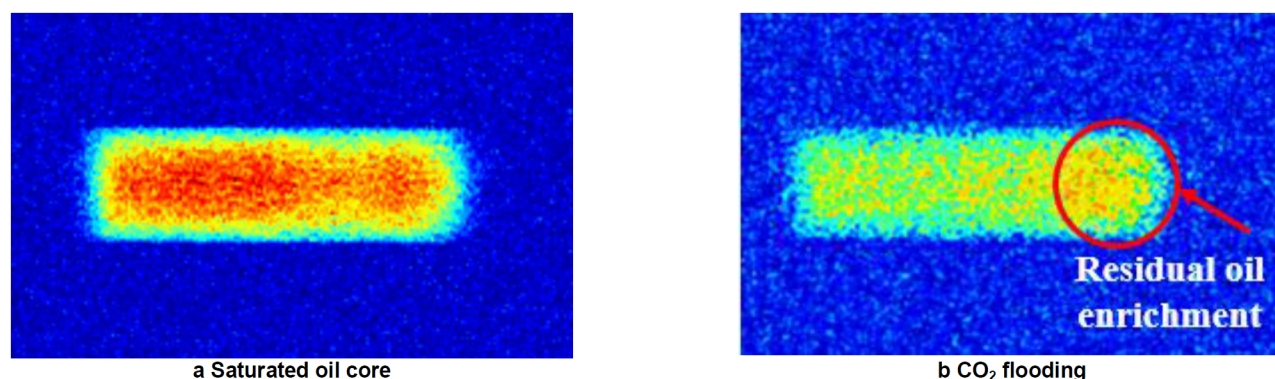


Figure 15—CO₂ flooding magnetic resonance imaging

Conclusion

The research employs online nuclear magnetic resonance to conduct CO₂ enhanced reverse hydraulic fracturing assisted oil displacement experiments, identifying the contribution of different stages to recovery and CO₂ sequestration ratio, quantifying the degree of crude oil mobilization in different pores, and revealing the mechanism for improving recovery. The main conclusions are as follows.

1. The oil recovery of CO₂ enhanced reverse hydraulic fracturing assister oil displacement is 88.8%, including 12.3% from fracturing fluid imbibition during the shut-in stage, 73.2% from CO₂ flooding, and 3.3% from dissolved gas drive.
2. The CO₂ enhanced reverse hydraulic assisted oil displacement sequestration is 65.5%, the CO₂ sequestration is 57.6%, which is an increase of 7.9%.
3. Compared with water injection huff and puff, CO₂ injection huff and puff, and CO₂ flooding, CO₂ enhanced reverse hydraulic assisted oil displacement increases recovery range from 16.7% to 75%.

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